See instructions in attached doc.

https://gemini.google.com/app/0858ee74070bc1bb

User prompt: See instructions in attached doc.

Response: **Verdict:** Do not approve revision 3; its recommendation to abandon all forward studies and stop the program rests on flawed power derivations in Section 2 and Section 4, while its redesigned Study 1 introduces a fatal inter-market dependence bug that invalidates Wald's sequential sign test. Specifically, Section 4's forward power table understates power by forcing all weekly purchases in `account()` to lock up cash without settling until week's end (shrinking the 60th bet of a week to 0.24% of cash), by solving for terminal wealth $\mathbb{P}(W_N \geq 20) = 0.80$ ($N = 11.8/e^2$) instead of the stopping boundary $\mathbb{P}(\sup_{n \leq N} W_n \geq 20) = 0.80$ ($N = 8.67/e^2$), and by retaining the gate's 100σ worst-loss cap ($L = 2$ against $\sigma = 0.02$) when a 10σ step stop ($L = 0.20$) cuts the two-year 80%-power Sharpe requirement on certified Hyperliquid data from 4.0 to 2.09. Meanwhile, Study 1's new twin-market plant measure evaluates all 100 study markets' picks on the *same* K fixed fresh market seeds and the *same* noisy reference denominator, coupling all 100 classifications to a single finite sample draw and breaking independence across markets in Wald's SPRT, while non-linear bankruptcy resurrection in the frozen kernel creates positive convexity payoffs under planted drift that do not cancel between twins.

Ranked Findings (Part 1 of 2)

1. Critical — Study 1's shared-K twin-market measure breaks independence across the 100 study markets, fails to cancel non-linear kernel defects, and understates classification error

- **Where:** `architecture.md` §5.1: *"A pick earns from the plant when the plant's effect on its score is at least half the plant's effect on the locked reference program's score. The effect is the program's mean score over `K` fresh planted markets minus its mean score over their twins, which share every seed and have plant strength zero. The fresh markets' seeds are fixed in the manifest and shared by all picks… Whatever the frozen kernel pays that is not the plant, a defect or a leak, it pays on both twins, and the difference removes it."* Also §5.2 (*"Dependence between markets"*), §5.5 Wall 1 (*"misclassified with probability 3·10⁻⁵"*), and `arithmetic.py.txt` line 131 (`norm.sf(4)`).

- **Construction:**

  1. **Sharing K fresh markets across all picks breaks inter-market independence in Wald's SPRT:** Let $\mathcal{E}_{\text{fresh}} = \{(M_k(\alpha), M_k(0))\}_{k=1}^{K}$ be the K fixed fresh twin-market pairs shared by all picks. On each study market $m \in \{1, \ldots, 100\}$, arm $a \in \{A, B\}$ selects a pick $\pi_{a,m}$, which is classified as earning from the plant if

$$\hat{\Delta}(\pi_{a,m}; \mathcal{E}_{\text{fresh}}) \geq \frac{1}{2}\hat{\Delta}(\pi_{\text{ref}}; \mathcal{E}_{\text{fresh}}).$$

Because every study market $m = 1, \ldots, 100$ is judged against the *same* realization of $\mathcal{E}_{\text{fresh}}$ and the *same* sample threshold $\frac{1}{2}\hat{\Delta}(\pi_{\text{ref}}; \mathcal{E}_{\text{fresh}})$, the discordant indicators $D_1, \ldots, D_{100}$ are **not independent** across m: they share the common random environment $\mathcal{E}_{\text{fresh}}$. If the finite draw $\mathcal{E}_{\text{fresh}}$ happens to yield a slightly low sample estimate $\hat{\Delta}(\pi_{\text{ref}}; \mathcal{E}_{\text{fresh}}) < \Delta_{\text{ref}}$ or favors the smooth funding family named in the agent's brief over G100's trees, it shifts the win probability on all 100 study markets simultaneously, violating the i.i.d./independent-increment requirement of `sign_test()`.

2. **Kernel defects (bankruptcy resurrection) and lookahead leaks do not cancel between twins:** Cancellation requires score additivity $S(\pi, M_k(\alpha)) = S_{\text{plant}}(\pi, \alpha) + S_{\text{defect}}(\pi, M_k(0))$. In `gate/markets.py`, adding planted drift/funding $\alpha > 0$ changes the return series $r_t(\alpha) = r_t(0) + \alpha s_t$, causing the cumulative price and equity paths on $M_k(\alpha)$ and $M_k(0)$ to diverge. Because the frozen kernel's bankruptcy resurrection bug (Paper Finding 2c) and margin liquidations are non-linear threshold functions of equity ($W_t \leq 0$), they do not affect both twins equally. Specifically, because bankruptcy truncates left-tail losses at -100% while resurrection preserves post-bankruptcy upside, adding variance/drift from the plant $\alpha > 0$ to a high-leverage noisy pick creates a strictly positive Jensen's inequality (call-option convexity) gain:

$$\mathbb{E}\big[\max(-L, R_{\text{twin}} + R_{\text{plant}})\big] - \mathbb{E}\big[\max(-L, R_{\text{twin}})\big] > 0$$

even for a degenerate strategy with zero forecasting edge. Similarly, any pick that multiplies an evaluator leak by the funding input (`pos = funding * leak`) evaluates to near-zero on the zero-plant twin and non-zero on the planted market, multiplying rather than canceling the leak.

3. **The $4\,\text{SE}$ / $3 \cdot 10^{-5}$ classification error bound is mathematically false:** Section 5.5 sets K so that $\frac{1}{2}\hat{\Delta}_{\text{ref}} \geq 4\,\text{SE}(\hat{\Delta}(\pi))$ on calibration picks and claims misclassification probability $\Phi(-4) = 3.17 \times 10^{-5}$. First, this treats the threshold $\frac{1}{2}\hat{\Delta}_{\text{ref}}$ as deterministic; for an uncorrelated pick with $\text{SE}(\hat{\Delta}(\pi)) = \text{SE}(\hat{\Delta}_{\text{ref}})$, the standard error of $\hat{\Delta}(\pi) - \frac{1}{2}\hat{\Delta}_{\text{ref}}$ is $\quad 1 + 0.25\,\text{SE} = 1.118\,\text{SE}$, reducing the margin to $3.58\,\text{SE}$ (1.7×10^{-4}, Over $5\times$ larger). Second, K is calibrated only on G100's calibration picks, which does not bound the variance of agent picks during the study. Third, $3 \cdot 10^{-5}$ applies only if every pick's true effect is exactly 0 or Δ_{ref}; picks that partially capture the plant with true effect $\Delta(\pi) \in (0.35\Delta_{\text{ref}}, 0.65\Delta_{\text{ref}})$ sit within 0–$1.2\,\text{SE}$ of the threshold and are misclassified up to 50% of the time, with errors correlated across all 100 study markets via the shared K seeds.

- **Outcome:** Invalidates Study 1's sequential sign test across the 100 markets, allows bankruptcy-resurrection convexity to masquerade as plant discovery, and understates pick misclassification rates by orders of magnitude.

- **Severity:** Critical.

2. Critical — Section 4's negative conclusion is falsified by three compounding errors: intra-week cash lockup in `account()`, terminal instead of stopping power in `11.8 / e²`, and the gate's artificial 100σ loss cap ($L = 2$)

- **Where:** `architecture.md` §1: *"an exact test needs about `11.8 / e²` independent bets for 80% power at a net edge `e`"* ; §2.2: *"The gate has `L = 2` against 2% a four-hour step… Capped, 80% power over 546 steps needs Sharpe 27.7 (26.0 at even odds), and over two years 4.0."* ; §4 Table & text: *"No forward study of a reading agent has honest power on a certified venue within two years… at the plausible ceiling of 0.05 the smallest count with 80% power is above 60 independent bets a week, sustained for two years, and under the null the same simulation crosses 20 in 1% of paths."* Also `arithmetic.py.txt` `capped_bet()`, `betting_cost()`, `account()`, `forward_power()`.

- **Construction:**

  1. **Artificial intra-week cash depletion in `account()`:** In `arithmetic.py.txt` (lines 145–149), `account()` places all `bets` for an entire week inside `for _ in range(bets): stake = fraction * cash; cash = cash - stake` before adding *any* settlement (`cash = cash + payout`). Thus, within each week, the k-th purchase stakes $f(1-f)^{k-1}$ of initial weekly cash. At $m = 60$ bets/week and $f = 0.05$, the 60th bet stakes only $0.05(0.95)^{59} = 0.24\%$ of weekly cash (an average stake of $\frac{1-0.95^{60}}{60} = 1.59\%$ instead of 5%, cutting capital deployment by $3.1\times$). When short-horizon perp theses or daily events settle sequentially across the week (so at most 5–8 positions are simultaneously open), 60 bets/week reaches 80% power at $e = 0.044$ (below Fable's 0.05 ceiling, vs. 0.055 in §4), and two-year power at $e = 0.04$ rises from 0.50 to 0.73.

  2. **Conflating terminal wealth $\mathbb{P}(W_N \geq 20)$ with stopping power $\mathbb{P}(\sup_{n \leq N} W_n \geq 20)$ in `11.8 / e²`:** In `betting_cost()`, Fable solves $\frac{1}{2}u^2 - z_{0.80}u = \ln 20 \implies u = 3.430, u^2 = 11.765$ (`11.8 / e²`), which is the probability that *terminal* wealth at step N exceeds 20. Because the account stops as soon as wealth hits 20 ($\sup_{n \leq N} W_n \geq 20$), applying the exact Bachelier–Lévy reflection formula for the running maximum of drift-diffusion $\ln W_t$ with drift $e^2/2$ and volatility e:

$$\mathbb{P}\left(\sup_{0 \leq t \leq N} W_t \geq 20\right) = \Phi\left(\frac{u}{2} - \frac{\ln 20}{u}\right) + 20\,\Phi\left(-\frac{u}{2} - \frac{\ln 20}{u}\right) = 0.80$$

yields $u = 2.945 \implies N = \frac{8.67}{e^2}$ (**26.3% fewer bets** than $11.765/e^2$; at $u^2 = 11.765$, true stopping power is 89.9%, not 80%). Moreover, Fable notes that `account()` crosses 20 in only 1% of null paths by week 104, meaning the fixed-fraction test wastes 80% of its $\alpha = 0.05$ Ville budget on $t > 104$ weeks.

3. **Constructive forward designs on certified sources (Hyperliquid + public filings) with $\geq 80\%$ power in ≤ 2 years:**

   - *Systematic / hybrid perp track (Q3):* Section 4 claims the gate's exact test requires **Sharpe 4.0** over two years. That is caused solely by setting the worst-loss bound to $L = 2.0$ (200% loss = 100σ against 4-hour $\sigma = 0.02$), which caps the betting fraction at $\lambda = 1/L = 0.50$ when optimal Kelly at Sharpe 2.0 is $\lambda^* = \frac{\mu}{\sigma^2} = \frac{2.0/\ \ 2190}{0.02} = 2.14$. Enforcing a -20% (10σ, $L = 0.20$) per-step stop or trading bounded cross-sectional spreads sets $1/L = 5.0 > \lambda^*$, removing the cap and reaching **80% exact Ville stopping power over two years at annualized Sharpe** $S = \ \ 8.67/2 = 2.08$ (or $S = 2.43$ terminal)—nearly halving the 4.0 wall.

   - *Reading-agent perp & filing track (Q2 & Q3):* Hyperliquid's certified API provides 150+ main-dex and HIP-3 perps with 4-hour/hourly funding alongside certified public macro/SEC filing feeds. Scoring 6.5 independent catalyst or cross-sectional relative-value pairs per day (45 bounded bets/week, $N = 4,680$ in two years) with step loss bounded at L so Kelly is uncapped reaches **80% exact stopping power at net edge** $e = \ \ 8.67/4680 = 0.0430$ (52.15% hit rate, well below Fable's 0.05 ceiling). Furthermore, §4 falsely dismisses Q2 by claiming a no-search baseline lacks current data: pairing Arm A (agent with certified text filings + current Hyperliquid prices/funding) against Arm B (same agent with the *same* current Hyperliquid prices/funding up to timestamp t, minus text filings) cancels market return variance σ^2_{mkt} on identical timestamps and reaches 80% power in under 6 months.

- **Outcome:** Overturns the core negative conclusion in §1, §4, §6.1, and §11.1 that no forward study on certified venues can achieve 80% power within two years at plausible edges (≤ 0.05 or Sharpe ≈ 2).

- **Severity:** Critical.

3. Critical — Study 1's claim of "92% power at the target under every dependence" is false across its own $[0.15, 0.25]$ calibration band, and `arithmetic.py.txt` hides failing rows from §5.3's table

- **Where:** `architecture.md` §5.3: *"Difficulty is one plant strength, chosen without a model on 400 calibration markets a grid value: the one at which G100's rate is nearest 0.2, inside*

0.15 to 0.25 (standard error 0.02)." ; §11.4: *"Study 1 at up to 100 markets in place of 60… I recommend yes: it holds 92% power at the target under every dependence between the arms."* ; §8 Finding 4: *"the lower boundary claims nothing, so its (0.25, 0.20) case is a true rejection"*. Also `arithmetic.py.txt` `study1()` lines 102–125.

- **Construction:**

  1. **Contradiction between §11.4's "92% power" claim and §5.3's calibration band:** Section 5.3 permits any calibration grid strength where G100's rate is in $[0.15, 0.25]$. In §5.3's own table, at `(0.5, 0.25)` under least overlap, power is **0.758 (75.8%)** with a **9.9% futility stop rate** and **14.3% undecided rate**—not 92%. Furthermore, because G100's calibration rate is estimated on 400 markets with standard error $0.25 \times 0.75/400 = 0.0217$, a grid strength whose sample rate is $\hat{p}_B = 0.24$–0.25 has a true G100 rate of $p_B = 0.30$ within 2.3σ. In `arithmetic.py.txt` lines 112–114, Fable computed `("0.5, 0.3, least overlap", 0.5, 0.3, 0.0)` and `("0.5, 0.3, both 0.05 (Argon)", 0.5, 0.3, 0.05)` under the new $\theta_1 = 0.7, n_{\max} = 100$ test—where true $\theta = \frac{0.5}{0.8} = 0.625 < 0.70$, **power collapses to 0.508 (50.8%)**, futility stops jump to **20.0%**, and undecided hits **29.2%**—but **omitted those rows from §5.3's table**.

  2. **The $(0.25, 0.20)$ positive-overlap case (`both = 0.1875`) fails to reject 67.9% of the time:** In §8 Finding 4, Fable disposes of our $(0.25, 0.20)$ positive-correlation example by saying *"its (0.25, 0.20) case is a true rejection"* and includes `("0.25, 0.2, both 0.1875 (Argon)", 0.25, 0.2, 0.1875)` in `arithmetic.py.txt` line 118, again omitting the printed output from §5.3's table. In that row, $p_{10} = 0.0625$ and $p_{01} = 0.0125$ ($\theta = 0.833 > 0.70$), so $p_{\text{disc}} = 0.075$ yields only 7.5 expected discordant markets across all 100 trials. Because crossing `upper = ln(19) = 2.9444` with step size `win = ln(1.4) = 0.3365` requires at least $\lceil 2.9444/0.3365 \rceil = 9$ discordant wins, `sign_test(100, 0.7, ...)` leaves the test **undecided 67.5% of the time** (claiming only **32.1%** of the time) despite $\theta = 0.833 > 0.70$.

- **Outcome:** Falsifies §11.4's assurance to the human that 100 markets guarantees 92% power across dependences and hides the $(0.5, 0.3)$ and $(0.25, 0.20)$ rows computed by `arithmetic.py.txt` from §5.3's decision table.

- **Severity:** Critical.

CONTINUE FOR PART 2

User prompt: continue

Response: Ranked Findings (Part 2 of 2)

4. High — Section 6.3 and Section 8's disposition of Finding 5 fail under outcome-dependent recording delays across dependent contracts, and `late_records(20)` (1.4% vs. 3.7%) is an artifact of capital lockup and horizon truncation

- **Where:** `architecture.md` §6.3: *"Cash plus the conditional expected payout of every position not yet recorded as paid is a non-negative supermartingale on the record's own filtration, and cash never exceeds it. A payout recorded late, or never, is a position still open in that sum. No assumption is made about when records arrive: a delay may depend on outcomes already realized, because every stake is fixed before its own outcome, and a null stated on a finer filtration implies the one on the record. So the 5% bound holds under any dependence between contracts and any recording failure"*. Also §8 Finding 5 (*"under a fair null recorded cash reaches 20 in 3.7% of paths on time and 1.4% with every payout 20 purchases late"*) and `arithmetic.py.txt` lines 192–207 (`late_records`).

- **Construction:**

  1. **Why the tower-property argument in §6.3 fails under outcome-dependent delays and dependent contracts:** Let $\mathcal{F}_t^{\text{rec}}$ be the record's filtration and $\mathcal{G}_t$ be the world filtration at time t. Section 6.3 asserts that because a null on a finer filtration implies the one on the record ($\mathbb{E}[x_j \mid \mathcal{G}_{\tau_j}] \le c_j \implies \mathbb{E}[x_j \mid \mathcal{F}^{\text{rec}}_{\tau_j}] \le c_j$), the process

     $$V_t = C_t + \sum_{i \in U(t)} \frac{s_i}{c_i} \mathbb{E}[x_i \mid \mathcal{F}_t^{\text{rec}}]$$

     is a supermartingale on $\mathcal{F}_t^{\text{rec}}$ under *"any dependence between contracts and any recording failure"*. This fails for two distinct reasons:

     - **Supermartingale failure of $\mathbb{E}[x_i \mid \mathcal{F}_t^{\text{rec}}]$ under outcome-dependent delays:** Suppose position 1 is registered at $\tau_1 = 1$ ($c_1 = 0.50$) and settles in the real world at $t = 2$, where losing records ($x_1 = 0$) are recorded immediately at $t = 2$ while winning records ($x_1 = 1$) are delayed until $t = 10$ (e.g., manual settlement review or schema change on winning payouts). Then the event $\{1 \in U(2)\}$ is $\mathcal{F}_2^{\text{rec}}$-measurable and implies $x_1 = 1$, so $\mathbb{E}[x_1 \mid \mathcal{F}_2^{\text{rec}}] = 1$. Now suppose position 2 is registered at $\tau_2 = 1.5$ on a negatively correlated contract (e.g., $x_2 = 1 - x_1$, $c_2 = 0.50$). At registration $\tau_2 = 1.5$, $\mathbb{E}[x_2 \mid \mathcal{G}_{1.5}] = 0.50 \le c_2$. But at $t = 2$, revealing $\{1 \in U(2)\}$ in $\mathcal{F}_2^{\text{rec}}$ simultaneously determines $x_2 = 0$, whereas if x_2's own settlement is also delayed, conditioning on the joint recording indicator $U(t)$ alters conditional expectations across steps.

- **Violation of $\mathcal{F}^{\text{rec}}_{\tau_j}$-predictability for a reading agent:** Because the agent reads live external information in $\mathcal{G}_{\tau_j}$ (news, prices, and real-world outcomes of already-settled events whose payout records are delayed in $\mathcal{F}^{\text{rec}}_{\tau_j}$), the agent's choice of whether to register purchase j, its target contract, and its registered stake s_j are $\mathcal{G}_{\tau_j}$-measurable, **not** $\mathcal{F}^{\text{rec}}_{\tau_j}$-measurable. If an agent observes in the news that its unrecorded position i won ($x_i = 1$) while i is still unrecorded in $\mathcal{F}^{\text{rec}}_{\tau_j}$, or selects contract j conditioned on x_i, then $\mathcal{F}^{\text{rec}}_{\tau_j}$ does not contain the agent's decision filtration.

2. **Why `late_records(20)` in `arithmetic.py.txt` gets `1.4%` vs. `3.7%`:** In `late_records(hold, bets=400)` (lines 192–207), `late_records(0)` settles every bet immediately before the next bet, so 100% of wealth is available to stake on all 400 sequential bets. In `late_records(20)`, holding *every* payout for 20 purchases locks up $1 - 0.95^{20} = 64.2\%$ of initial cash in the `pending` buffer (cutting active cash C_t to roughly half of total wealth on every bet) and **discards the final 20 pending payouts at `bet = 399` without ever adding them to `cash`**. Thus, `late_records(20)` reduces the crossing rate from 3.7% to 1.4% only because it cuts effective capital deployment and truncates the last 20 settlements—not because delayed recording is conservative. In fact, running `late_records(0)` and `late_records(20)` on the exact same uniform draws `u = rng.random((200_000, 400))` produces hundreds of paths where `late_records(20)` crosses 20 while `late_records(0)` never reaches 20 (the exact mechanism of Finding 5.1, 20.29 vs. 8.10).

- **Outcome:** Invalidates §6.3's proof that the record filtration needs no assumption on polling independence when contracts are dependent or when a reading agent observes real-world outcomes before the recorder, and misinterprets capital starvation in `late_records(20)` as statistical conservatism.

- **Severity:** High.

5. High — Study 1's market-by-market null, deterministic pre-computed G100 baseline, and operational crash rule distort the capability claim and allow a 5-session outage to trigger a false futility stop

- **Where:** `architecture.md` §5.1: *"B, G100, with a seed fixed in the manifest… Null: on every study market, the agent's chance of a pick that earns from the plant is at most G100's."* ; §5.2: *"The null is stated market by market, so a change in the served model mid-study changes the claim's population and not the size… An agent session that crashes or evaluates nothing is a miss for the agent and is never replaced. This can only lower the agent's rate. G100's picks on all 100 study markets are computed and classified before the first session"* ; §7: *"Five consecutive sessions lost to one operational cause pause it; they still count."*

- **Construction:**

1. **Conditioned on each fixed study market m, G100's outcome is deterministic ($p_{B,m} \in \{0, 1\}$):** Because study market m's seed, G100's seed, and the K fresh twin seeds are all fixed in the manifest (and §5.2 pre-computes and classifies all 100 G100 picks before the first agent session), G100's indicator on market m is deterministic: $B_m \in \{0, 1\}$. Stated literally *"on every study market"* ($p_{A,m} \leq p_{B,m}$ for all $m = 1, \ldots, 100$), the null asserts that on the ≈ 80 markets where $B_m = 0$, the agent has **zero** probability of finding the plant ($p_{A,m} = 0$).

2. **Mid-study model changes can cause Study 1 to reject when the average or current model is worse than G100:** Because the market-by-market null is an intersection hypothesis $H_0 = \bigcap_{m=1}^{100} \{p_{A,m} \leq p_{B,m}\}$, its negation is merely $\exists m$: $p_{A,m} > p_{B,m}$—the exact logical flaw Fable conceded on Study 2's mirror in Finding 1. Furthermore, because Wald's sequential test runs in manifest order and stops as soon as the LLR crosses $\ln(19) = 2.944$, if Model v_1 achieves $p_A = 0.50$ vs. $p_B = 0.20$ on markets 1–35 (crossing the upper boundary at $m = 35$) or if a silent provider update at $m = 25$ degrades the model to $p_A = 0.05 < p_B = 0.20$ after banking an early lead of 8 wins to 0 losses ($\text{LLR} = 8 \ln(1.4) = 2.69$), a single lucky win on markets 26–100 pushes the LLR across 2.944 and claims capability for a degraded population whose overall mean $\bar{p}_A < \bar{p}_B$.

3. **Counting 5 consecutive operational crashes before pausing (§7) drives the SPRT into the lower futility boundary:** Under $\theta_1 = 0.70$, each discordant win adds $\ln(0.7/0.5) = +0.3365$ to the LLR, while each discordant loss adds $\ln(0.3/0.5) = -0.5108$ ($1.52\times$ larger in magnitude), with lower futility boundary $-\ln(19) = -2.9444$. Counting an operational crash as an agent miss ($A_m = 0$) does not merely "lower the agent's rate" in a fixed-sample sense: in a sequential test, if an API/sandbox failure crashes 5 consecutive sessions (§7's pause threshold) across a stretch where G100 succeeded on 3–4 markets and the running LLR was slightly negative (e.g., -0.90), those operational crashes inject -1.53 to -2.04 of LLR penalty, crossing -2.9444 and **permanently terminating Study 1 for futility** before the operational pause in §7 can take effect.

- **Outcome:** Allows operational outages of ≤ 5 sessions to falsely terminate Study 1 at the futility boundary and equates rejecting a pointwise intersection null ($\exists m$) with overall capability.

- **Severity:** High.

6. Medium — Numerical and unit inconsistencies between `architecture.md` and `arithmetic.py.txt`

- **Where:** `architecture.md` preamble: *"Run: `audit/program/arithmetic.py.txt`, which prints every number here."* ; §1 (*"costs of 3 to 5 cents"*); §4 Table Row 2 (*"0.183, 0.115, 0.090 at 5, 10, 15: at an even mid, the event 12.4, 8.8 or 7.5 points likelier than the mid"*); §4 text (*"under the null the same simulation crosses 20 in 1% of paths"*); §5.3 (*"At about 22 sessions a week the study takes under five weeks at most and about two at the target"*); `arithmetic.py.txt` `forward_power()`, `event_costs()`, `rates()`.

- **Construction:**

  1. **Incompatible cost parameter between `forward_power()` ($c = 0.50$) and `event_costs()` ($c = 0.5275$) in §4 Table Row 2:** In `forward_power()` (line 168), `arithmetic.py.txt` solves for the 80%-power net edges `0.183, 0.115, 0.090` by calling `account(bets, 0.5, e, min(STAKE_CAP, kelly(0.5, e)))` with **`cost = 0.50`**. Then in `event_costs()` (line 183), it plugs those exact edges `0.183, 0.115, 0.090` into `100 * (cost * (1 + edge) - mid)` using **`cost = 0.527493`** (`mid = 0.50` plus 0.01 half-spread plus 0.017493 fee) to get `12.4, 8.8 or 7.5 points` above the mid. Because a binary contract at $c = 0.5275$ has lower return variance per unit of net edge than at $c = 0.50$ ($\frac{1-c}{c} = 0.896$ vs. 1.000) and higher Kelly fraction $\frac{ce}{1-c} = 1.116e$, solving `account()` consistently at `cost = 0.527493` yields required net edges of **0.172, 0.109, 0.085** (**11.8, 8.5, and 7.2 points** above the mid, not `12.4, 8.8, 7.5`).

  2. **Unit confusion in §1 ("costs of 3 to 5 cents"):** Section 1 states that event-contract exchanges have *"costs of 3 to 5 cents"*. In `event_costs()`, for `mid` $\in \{0.20, 0.50, 0.80\}$, the all-in cost above mid (1 cent half-spread plus Kalshi taker fee $0.07P(1-P)$) is **2.2 cents, 2.7 cents, and 2.1 cents per contract**—never 3 to 5 cents. Section 1 conflated `100 * fee / ask` (**3.4% to 5.5% of the stake** in §2.4) with cents per contract (at `mid = 0.20`, 5.5% of a 21-cent ask is 1.16 cents, not 5 cents).

  3. **False "same simulation" claim for the 1% null crossing rate in §4:** Section 4 states that every row in the table simulates staking *"Kelly's fraction for the true edge, capped at 5% of the cash left at that moment"* and claims that *"under the null the same simulation crosses 20 in 1% of paths"*. Under that simulation rule (`min(STAKE_CAP, kelly(0.5, edge))`), when `edge = 0.0`, `kelly(0.5, 0.0) = 0.0`, so the account stakes **0** and crosses 20 in **0% of paths**. Line 161 of `arithmetic.py.txt` obtained `0.010` (1%) by overriding the Kelly rule with a constant 5% stake `account(60, 0.5, 0.0, STAKE_CAP)` at `bets = 60` only.

  4. **Numbers in `architecture.md` absent from `arithmetic.py.txt`:** Despite the preamble's claim that `arithmetic.py.txt` prints every number in `architecture.md`, the script does not compute or print the Prophet Arena Brier numbers in §2.4 (`0.184 ± 0.006` vs. `0.187 ± 0.006`; instead `rates()` line 70 computes an unused 0.03 Brier gain, $10\times$ larger than Prophet Arena's 0.003

difference), the session throughput/duration figures in §5.3 (22 sessions/week, 5 weeks, 2 weeks), or the Stage 3 evaluation count in §5.5 (9 of 100).

- **Outcome:** Violates the reproducibility invariant that `arithmetic.py.txt` generates every number in the architecture document and overstates required probability points in §4's table.
- **Severity:** Medium.

7. Medium — Section 8's dispositions of Paper Findings 7, 10, and 11 rely on unhedged variance shares, a backwards Bonferroni CI direction, and an asymptotic $\hat{S} \to \infty$ DSR limit

- **Where:** `architecture.md` §8 Row 8: *"11: I withdraw 'martingale differences'; the understatement depends on funding's share of a step's variance, 1.002 at a share of 2.5·10⁻⁵, 4.9 at Argon's 0.10… 7: the DSR's limit in its own example is z = 11.3, so there is no ceiling; its examples show a Sharpe failing, which is the test working. 10: the 0.05/99 is the suite's own rule (paper, line 645), not mine; the sentence now names it."* Also `arithmetic.py.txt` `argon_paper()` lines 223–237.
- **Construction:**
  1. **Paper Finding 11 (Stationary bootstrap variance understatement):** In `argon_paper()` line 235, Fable computes the variance understatement ratio $\frac{1-w+83w}{1-w+9.723w}$, which equals `1.002` only at $w = 2.5 \times 10^{-5}$ (an unhedged directional perp position with 2% 4-hour price volatility and 1 bp funding volatility). However, for any cross-sectional funding-arbitrage or market-neutral strategy (long high-funding perps, short low-funding perps—the exact family planted in Study 1) or during funding spikes, price variance cancels across legs and funding's variance share w rises to 0.05–0.25, where the stationary bootstrap ($L = 6$) understates the variance of the sample mean by $2.9\times$ **to** $6.9\times$ (inflating t-statistics by $1.7\times$ **to** $2.6\times$).
  2. **Paper Finding 10 (v3 locked suite tolerances):** Line 229 of `arithmetic.py.txt` computes `beta.ppf(1 - 0.05 / 99, x + 1, 20_000 - x) ⩽ tolerance`. Applying a 99-way Bonferroni correction to an **upper** confidence bound (`1 - 0.05/99 = 0.999495`) narrows the acceptance region (increasing the suite's false-failure rate under a true null rather than controlling familywise error across 99 checks, which requires a **lower** bound `beta.ppf(0.05 / 99, x, 20_000 - x + 1) ⩽ alpha`). Yet because `tolerance` was set well above nominal α (0.0625 vs. 0.05, 0.015 vs. 0.01, 0.01 vs. 0.005), `argon_paper()`'s own output (`chance = 0.69, 0.62, 0.65`) proves that a buggy test with true Type-I error inflated to 5.6%, 1.2% ($+20\%$), or 0.75% ($+50\%$) still passes the locked suite **62% to 69% of the time**.

3. **Paper Finding 7 (Deflated Sharpe Ratio screen):** Fable argues that because $\lim_{\hat{S}\to\infty} z(\hat{S}) = 2 \quad \frac{545}{17} = 11.3 > 1.645$ at kurtosis $\hat{\gamma}_4 = 18$, there is "no ceiling". An asymptotic limit as annualized Sharpe $\hat{S}_{\text{ann}} \to \infty$ is irrelevant when `argon_paper()` line 226 confirms that over the 91-day holdout ($T = 546$), **an annualized Sharpe of 3.5 fails at every kurtosis** $\hat{\gamma}_4 \geq 1$ ($z \leq 0.748 < 1.645$) and **an annualized Sharpe of 6.0 with** $\hat{\gamma}_3 = -4, \hat{\gamma}_4 = 18$ **also fails** ($z = 1.613 < 1.645$). Worse, in `dsr_z()` (line 214), whenever a positively skewed strategy satisfies $\hat{\gamma}_3\hat{S} > 1 + \frac{\hat{\gamma}_4 - 1}{4}\hat{S}^2$ (e.g., $\hat{\gamma}_3 = 4, \hat{\gamma}_4 = 10, \hat{S} = 0.5$), the Mertens radicand `1 - skew * sharpe + (kurtosis - 1) / 4 * sharpe**2` becomes **negative**, crashing the DSR evaluator with a domain error.

- **Outcome:** Understates the bootstrap defect for market-neutral/funding strategies, leaves the v3 test suite blind to 20%–50% Type-I error inflation 62%–69% of the time, and overlooks finite-sample DSR rejection of Sharpe 3.5–6.0 strategies and positive-skewness domain crashes.
- **Severity:** Medium.

Summary Table of Ranked Findings

Rank	Severity	Target	Core Theoretical / Numerical Break
1	**Critical**	Study 1 twin-market measure & K rule (§5.1, §5.2, §5.5)	Sharing the same K fixed fresh market seeds and noisy reference estimate $\frac{1}{2}\hat{\Delta}_{\text{ref}}$ across all 100 study markets makes the discordant indicators $D_1, \ldots, D_{100}$ statistically dependent (invalidating Wald's SPRT); non-linear bankruptcy resurrection creates positive convexity under planted drift that does not cancel between twins; $3 \cdot 10^{-5}$ error ignores denominator variance and intermediate picks
2	**Critical**	Negative conclusion & Section 4 power table (§1, §2.2, §4, `account`, `betting_cost`)	`account()` locks up cash across all weekly bets before settling (shrinking the 60th bet to 0.24% of cash); `11.8/e²` computes terminal wealth $\mathbb{P}(W_N \geq 20)$ instead of stopping power $\mathbb{P}(\sup_{n\leq N} W_n \geq 20)$ ($8.67/e^2$, 26.3% fewer bets); replacing the 100σ gate cap ($L = 2$) with a 10σ step stop ($L = 0.20$) cuts 2-year 80%-power Sharpe on certified

Rank	Severity	Target	Core Theoretical / Numerical Break
			Hyperliquid data from 4.0 to 2.08, and 45 bounded perp/filing bets/week reaches 80% power at $e = 0.043$
3	**Critical**	Study 1 power across calibration band & dependences (§5.3, §8, §11.4, `study1`)	Falsely assures the human of "92% power at the target under every dependence" (§11.4) when §5.3's own $[0.15, 0.25]$ band drops to 75.8% at 0.25 and 50.8% at 0.30 (1.1σ from 0.25); `arithmetic.py.txt` computes the failing $(0.5, 0.3)$ and $(0.25, 0.20)$ rows (67.5% undecided despite $\theta = 0.833$) but omits them from §5.3's table
4	**High**	Section 6.3 validity proof & Finding 5 disposition (§6.3, §8, `late_records`)	Conditioning on $\mathcal{F}_t^{\text{rec}}$ fails when recording delays depend on realized outcomes across dependent contracts or when a reading agent observes real-world settlements before the recorder; `late_records(20)` gets 1.4% vs. 3.7% only because it locks up 64% of cash in `pending` and drops the last 20 payouts at step 400
5	**High**	Study 1 null, pre-computed G100, and crash rule (§5.1, §5.2, §7)	Pre-computing G100 on fixed seeds makes $B_m \in \{0, 1\}$ deterministic per market; market-by-market intersection null ($\forall m$) can reject under mid-study model changes when average capability is worse; counting 5 consecutive operational crashes before pausing (§7) injects up to -2.55 LLR, triggering a false futility termination
6	**Medium**	Script reproducibility & unit mismatches (`arithmetic.py.txt`, §1, §2.4, §4, §5.3)	`forward_power()` solves for edges at $c = 0.50$ while `event_costs()` applies them to $c = 0.5275$ (overstating required edge as $12.4, 8.8, 7.5$ pts vs. $11.8, 8.5, 7.2$ pts); §1 conflates 3.4%–5.5% of stake with "3 to 5 cents" (2.1–2.7 cents actual); §4 claims 1% null size under Kelly when `kelly(0.5, 0) = 0`; several text numbers are absent from the script

Rank	Severity	Target	Core Theoretical / Numerical Break
7	**Medium**	Dispositions of Paper Findings 7, 10, and 11 (§8, `argon_paper`)	Finding 11's `1.002` factor assumes an unhedged directional position ($w = 2.5 \cdot 10^{-5}$) rather than a market-neutral funding strategy ($2.9\times$ – $6.9\times$ variance understatement); Finding 10's upper-bound Bonferroni still lets 20%–50% inflated Type-I error pass 62%–69% of the time; Finding 7 ignores finite-sample rejection of Sharpe 3.5–6.0 at $T = 546$ and negative-radicand crashes for positive skewness

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